

Rethinking the policy optimization with Gradient Estimators

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Abstract

Permissive policy optimization can learn rapidly from rollout data and then fail abruptly after many successful updates. Policy-deviation bounds explain why a large change can be unsafe, but they do not say when a finite rollout stops supporting an accurate update for the current policy. We study this transition through two ordered questions. First, when is the ordinary full-trajectory importance-sampling (IS) estimator still reliable? We show that normalized effective sample size (ESS) controls the policy-mismatch component of its finite-sample error and, through smoothness, yields a high-probability one-step improvement guarantee. Second, when does coefficient control provide a better update? Lower gradient-estimation MSE is not enough: the controlled estimator must also retain sufficient alignment with the population gradient. We capture this requirement with a full fixed-step improvement certificate. An exactly solvable Gaussian bandit shows that the two criteria can switch at different ESS levels: PPO can have lower MSE while IS still gives the larger exact expected improvement. Along the example’s monotone off-policy path, the safe certificate oracle—IS, PPO, or no update—reduces exactly to two ESS thresholds. In the reported large-model trajectories, ESS-conditioned safeguards preserve early learning while preventing late failure.

1 Introduction

Reinforcement learning with verifiable rewards repeatedly collects responses from a rollout policy and reuses each batch for one or more policy updates. A permissive update can work surprisingly well in this setting: performance rises quickly and remains stable, but then collapses abruptly. Figure 1 shows this delayed failure. Rollout data support useful learning for many updates before the same update rule becomes unstable.

Classical theory explains why moving too far from the rollout policy can be dangerous [11, 13, 14]. That explanation is essential, but it does not explain why failure is delayed. A policy-deviation bound controls a population approximation; it does not track how much information a finite rollout still provides for the gradient of the current policy. The practical question raised by Figure 1 is therefore statistical: when does the rollout cease to support a reliable next update?

Our starting point is that nominal batch size can remain fixed while effective support contracts. Importance weights become increasingly uneven as the current policy moves away from the rollout policy, so fewer responses determine the update. Normalized ESS measures this contraction. We connect it first to gradient reliability and then to policy improvement.

This separation leads to two ordered questions. First, when is the IS importance-weighted gradient reliable enough to improve the population objective? Second, if IS becomes unreliable, when is a controlled estimator actually better? The second question cannot be answered by ESS or MSE alone. Clipping may reduce variance and MSE while also removing useful population-gradient alignment. We therefore compare candidate estimators using the full fixed-step improvement certificate.

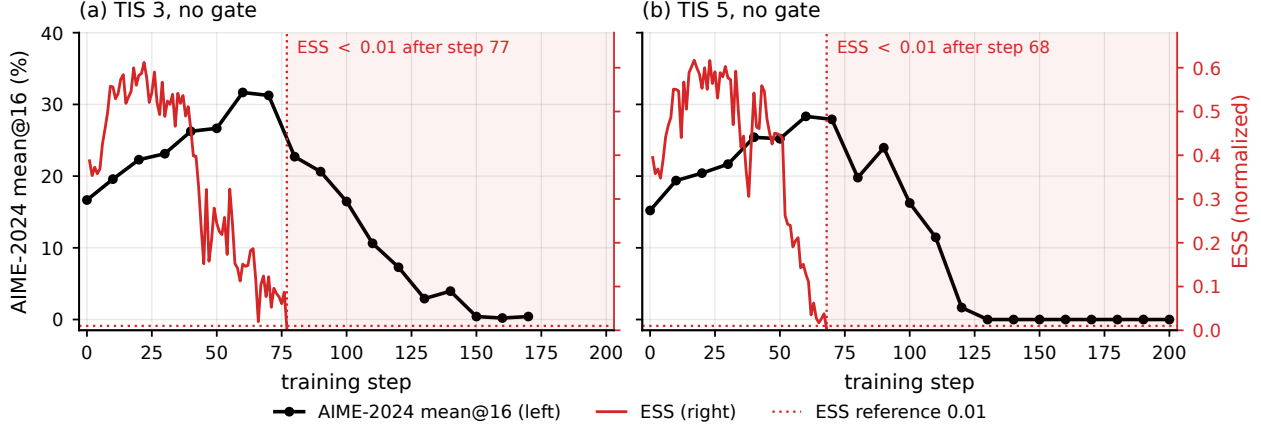


Figure 1: A delayed failure in Qwen3-30B-A3B training. The permissive runs improve for many updates and fail only after the effective support of the rollout data has sharply deteriorated. This transition is the phenomenon studied in the paper.

An exactly solvable Gaussian bandit makes the distinction concrete. Every population moment and the expected one-step gain are available without Monte Carlo error, allowing us to separate the MSE crossover from the actual improvement crossover. We then test the resulting regime prediction in large-model training. These experiments use ESS as a diagnostic of effective support and test the predicted support-to-intervention ordering.

1.1 Contributions

1. We show how normalized ESS controls the finite-sample error of the IS gradient estimator from a fixed rollout and yields a one-step population improvement guarantee.
2. We distinguish estimator accuracy from update quality and derive the fixed-step condition under which coefficient control has the stronger improvement certificate.
3. We separate the two crossovers exactly in a Gaussian bandit and then evaluate ESS-conditioned safeguards in large-model training.

2 Related work

Policy-deviation guarantees. Trust-region and proximal methods control policy movement to obtain stable improvement guarantees or practical update rules [11, 13, 14]. These analyses provide the standard explanation for why large policy changes can be unsafe. Our framework is complementary: it studies when a fixed finite rollout ceases to provide a reliable estimate of the current policy gradient. The distinction is between a population-level approximation guarantee and a finite-sample reliability question.

Importance sampling and effective support. Importance-sampling policy optimization uses likelihood-ratio moments and Rényi divergence to quantify the reliability of off-policy estimators [5, 6]. In these results, effective rather than nominal sample size controls estimation error. We apply the same principle directly to the vector-valued policy-gradient estimator and use normalized ESS as the regime coordinate.

Reliable policy-gradient updates. Smoothness-based analyses translate the error of a finite gradient estimate into a lower bound on policy improvement and use this relation to choose step sizes or sample sizes [8, 10]. We combine this optimization argument with an ESS-dependent off-policy reliability bound.

Stability in RLVR. Recent methods alter ratio boundaries, sequence weights, divergence controls, or update rules to improve the stability of large-model reinforcement learning [2, 3, 7, 9, 16, 18, 19]. Rather than proposing another static boundary, we study the statistical transition that makes a permissive update unreliable and then ask when an intervention can recover reliability.

3 Preliminaries

This section provides the background and notation used in the following analysis.

3.1 Policy-gradient estimators from a fixed rollout

Problem formulation. Let $X \sim \nu$ denote a context, $Q(\cdot | X)$ the rollout policy, and $P_\theta(\cdot | X)$ the current policy. A complete sample is $Z = (X, Y)$, where $Y = (Y_1, \dots, Y_T)$ and $R(Z)$ is its reward. For each factor t , define the likelihood ratio and score

$$r_{\theta,t}(Z) := \frac{P_\theta(Y_t | X, Y_{<t})}{Q(Y_t | X, Y_{<t})}, \quad \ell_{\theta,t}(Z) := \nabla_\theta \log P_\theta(Y_t | X, Y_{<t}). \quad (1)$$

Here t indexes a policy factor, such as a token or decision step. Because both policies factorize along the trajectory, the complete-sample likelihood ratio is

$$W_\theta(Z) := \frac{P_\theta(Y | X)}{Q(Y | X)} = \prod_{t=1}^T r_{\theta,t}(Z). \quad (2)$$

The learning objective is to maximize the expected reward $J(\theta) := \mathbb{E}_{P_\theta}[R]$. Let $A(Z) := R(Z) - b(X)$ be a baseline-centered reward, where $b(X)$ does not depend on Y . Under $P_\theta(\cdot | X) \ll Q(\cdot | X)$, change of measure expresses the population gradient under the rollout distribution:

$$g(\theta) := \nabla J(\theta) = \mathbb{E}_Q[W_\theta f_\theta], \quad f_\theta(Z) := A(Z) \sum_{t=1}^T \ell_{\theta,t}(Z) = A(Z) \nabla_\theta \log P_\theta(Y | X). \quad (3)$$

While trust-region arguments motivate safeguards such as TRPO and PPO-style clipping [13, 14], these arguments are typically made at the population level. Practical algorithms, however, do not observe $g(\theta)$ and must rely on its finite-sample counterpart. Given i.i.d. rollout samples $Z_i \sim \nu(dx)Q(dy | x)$, $i = 1, \dots, N$, the ordinary full-trajectory importance-sampling (IS) policy-gradient estimator is

$$\hat{g}_N(\theta) := \frac{1}{N} \sum_{i=1}^N W_{\theta,i} f_{\theta,i}. \quad (4)$$

The importance-weighted nature of this estimator can produce poor agreement between the finite-sample update and the population gradient [5, 6]. This exposes an important but often overlooked question: when is the finite-sample gradient reliable?

Choice of gradient estimator. To formalize this comparison, write $\mathbf{r}_\theta = (r_{\theta,1}, \dots, r_{\theta,T})$ and let $u = (u_1, \dots, u_T)$ be a collection of coefficient rules, with $u_t : \mathbb{R}_+^T \times \mathbb{R} \rightarrow \mathbb{R}$. Define

$$\psi_\theta(u; Z) := A(Z) \sum_{t=1}^T u_t(\mathbf{r}_\theta, A(Z)) \ell_{\theta,t}(Z), \quad \hat{g}_N(u; \theta) := \frac{1}{N} \sum_{i=1}^N \psi_\theta(u; Z_i). \quad (5)$$

Existing ratio-based policy-gradient estimators can be represented by particular choices of u . The IS estimator sets $u_{\text{IS},t} = W_\theta$ for every factor. One popular alternative is PPO/GRPO ratio clipping, whose clipped surrogate induces the following gradient-masking rule with radius $0 < \epsilon < 1$ [14, 15]:

$$u_{\text{P},t}(\mathbf{r}, A) := r_t [\mathbf{1}\{A \geq 0, r_t \leq 1 + \epsilon\} + \mathbf{1}\{A < 0, r_t \geq 1 - \epsilon\}]. \quad (6)$$

Such clipping rules create a conventional bias–variance tradeoff. More aggressive clipping can reduce sampling variation, but it can also increase bias by removing gradient terms. This raises the natural question: what governs the reliability of a gradient estimator, and when is clipping worth the signal it removes? We answer the first part by introducing Rényi divergence and effective sample size (ESS).

3.2 Gradient estimators and Rényi divergence

The dispersion of the importance weights indicates how dissimilar P_θ and Q are. Rényi divergence formalizes this notion as an information-theoretic dissimilarity between probability measures [12, 17], while importance-sampling analyses connect it to the finite-sample behavior of importance-weighted means [5, 6].

Rényi divergence is a family of measures indexed by a moment order $\alpha > 1$. For the current and rollout policies, define

$$D_\alpha(P_\theta \| Q) := \frac{1}{\alpha - 1} \log \mathbb{E}_Q[W_\theta^\alpha], \quad d_\alpha(P_\theta \| Q) := \exp\{D_\alpha(P_\theta \| Q)\}. \quad (7)$$

Larger values of α place progressively more emphasis on rare, large likelihood ratios. The order-two case is central because $\mathbb{E}_Q[W_\theta^2]$ controls standard MSE bounds for bounded importance-weighted contributions. Importantly, its reciprocal is the normalized effective sample size [5, 6]:

$$\rho_\theta := \frac{1}{d_2(P_\theta \| Q)} = \frac{1}{\mathbb{E}_Q[W_\theta^2]} = \exp\{-D_2(P_\theta \| Q)\}. \quad (8)$$

When $\mathbb{E}_Q[W_\theta^2] < \infty$, Jensen’s inequality and $\mathbb{E}_Q[W_\theta] = 1$ give $\rho_\theta \in (0, 1]$, and $N\rho_\theta$ is the effective sample count corresponding to nominal batch size N . As the second moment of the likelihood ratio grows, both ρ_θ and this effective count contract. The identity $\mathbb{E}_Q[(W_\theta - 1)^2] = \rho_\theta^{-1} - 1$ shows that ρ_θ is exactly the inverse second-moment measure of ratio dispersion.

Importance-sampling theory makes this quantity operational. If $\|f_\theta\|_\infty := \sup_z \|f_\theta(z)\|_2 < \infty$, the MSE of the IS estimator is at most $\|f_\theta\|_\infty^2 / (N\rho_\theta)$. Normalized ESS therefore captures the policy-mismatch component of IS reliability. The next section determines when that reliability is sufficient for improvement and then asks whether coefficient control offers a better update.

4 Theoretical Analysis

This section answers the preceding questions in order. We first connect gradient-estimator reliability to policy improvement, then use normalized ESS to control the reliability of the IS estimator, and finally ask when clipping provides the stronger update.

4.1 Gradient-estimator reliability and policy improvement

The preceding section distinguishes the finite-sample IS estimate from the population gradient it targets. Here, we connect estimator reliability to a lower bound on policy improvement. The following lemma formalizes this connection for an arbitrary gradient estimator.

Lemma 1 (A reliable gradient estimate yields policy improvement). *Let $\hat{g}$ be an estimate of $g = \nabla J(\theta)$ satisfying $\|\hat{g} - g\|_2 \leq \varepsilon$. Suppose J is L -smooth along the update segment for $L > 0$, and define $\gamma = L^{-1}(1 - \varepsilon/\|\hat{g}\|_2)_+$, with $\gamma = 0$ when $\hat{g} = 0$. Then*

$$J(\theta + \gamma\hat{g}) - J(\theta) \geq \frac{1}{2L} (\|\hat{g}\|_2 - \varepsilon)_+^2. \quad (9)$$

This deterministic lemma uses the same smoothness mechanism as existing policy-gradient analyses [8, 10]. Its implication is direct: the error radius measures how far the finite-sample direction can lie from the population gradient. When this radius is smaller than $\|\hat{g}\|_2$, the estimator yields a positive lower bound on policy improvement.

ESS and the IS update. However, the population gradient is unknown, so this reliability cannot be evaluated directly. Normalized ESS provides the bridge: for the IS estimator, it converts likelihood-ratio dispersion into a bound on gradient-estimation error. At update k , condition on the preceding updates and write $W_k := W_{\theta_k}$, $f_k := f_{\theta_k}$, $g_k := g(\theta_k)$, $\hat{g}_k := \hat{g}_N(\theta_k)$, and $\rho_k := \rho_{\theta_k}$. Assume that the N rollout samples are independent draws from Q and that $\|f_k\|_\infty := \sup_z \|f_k(z)\|_2 < \infty$.

Because the IS estimator is unbiased, its exact mean-squared error is

$$\mathbb{E}\|\hat{g}_k - g_k\|_2^2 = \frac{1}{N} \{ \mathbb{E}_Q[W_k^2 \|f_k\|_2^2] - \|g_k\|_2^2 \}. \quad (10)$$

Using $\|f_k(z)\|_2 \leq \|f_k\|_\infty$ and $\rho_k^{-1} = \mathbb{E}_Q[W_k^2]$ gives the interpretable upper bound

$$\mathbb{E}\|\hat{g}_k - g_k\|_2^2 \leq \frac{\|f_k\|_\infty^2}{N\rho_k}. \quad (11)$$

Likelihood-ratio dispersion therefore enters through the effective count $N\rho_k$, replacing the nominal sample size N . Equation (11) controls the mean-squared error; the next lemma converts this result into a high-probability error radius for the realized update.

Lemma 2 (Normalized ESS controls the gradient-error radius). *Under the preceding setup, suppose $d_2(P_{\theta_k} \| Q) < \infty$ and $\|f_k\|_\infty < \infty$. Then, for any $0 < \delta \leq 1$, with probability at least $1 - \delta$,*

$$\|\hat{g}_k - g_k\|_2 \leq \frac{\|f_k\|_\infty}{\sqrt{\delta N \rho_k}}. \quad (12)$$

The radius grows as $(N\rho_k)^{-1/2}$. Thus, low ESS weakens the reliability guarantee for the IS gradient, consistent with the motivating example in the introduction. We now combine this radius with Lemma 1 to obtain an ESS-dependent improvement guarantee.

Theorem 3 (ESS-dependent improvement for the IS estimator). *Under the conditions of Lemma 2, suppose that J is L_k -smooth along the update segment for $L_k > 0$. For any $0 < \delta \leq 1$, define*

$$\varepsilon_k(\delta) := \frac{\|f_k\|_\infty}{\sqrt{\delta N \rho_k}}, \quad \gamma_k := \frac{1}{L_k} \left(1 - \frac{\varepsilon_k(\delta)}{\|\hat{g}_k\|_2} \right)_+, \quad (13)$$

with $\gamma_k = 0$ when $\hat{g}_k = 0$. Then, with probability at least $1 - \delta$,

$$J(\theta_k + \gamma_k \hat{g}_k) - J(\theta_k) \geq \frac{1}{2L_k} (\|\hat{g}_k\|_2 - \varepsilon_k(\delta))_+^2. \quad (14)$$

Theorem 3 makes the role of ESS explicit. Its lower bound is positive when

$$\|\hat{g}_k\|_2 > \frac{\|f_k\|_\infty}{\sqrt{\delta N \rho_k}}.$$

Thus, the IS update remains certifiably useful while effective support is sufficiently large relative to the observed gradient signal. Holding the other terms fixed, both the admissible step and the certified improvement shrink as ESS falls. This complements population policy-deviation guarantees, which characterize the consequences of changing the policy, by controlling the finite-rollout direction used to make that change. Having established when the IS update is reliable, we next ask when an alternative gradient estimator provides the stronger one-step certificate.

4.2 Choosing an alternative gradient estimator

The preceding result connects the reliability of the IS estimator to policy improvement. When this guarantee weakens, one may instead use a controlled coefficient rule, such as truncated IS or PPO-style masking. We compare a general rule u with IS in two steps: first as a question of gradient estimation, and then as a question of policy improvement at a common fixed step. Write $\hat{g}_k(u) := \hat{g}_N(u; \theta_k)$. For a rule u , adding and subtracting the estimator mean gives the MSE decomposition

$$\begin{aligned} m_k(u) &:= \mathbb{E} \|\hat{g}_k(u) - g_k\|_2^2 \\ &= \|\mathbb{E}[\hat{g}_k(u)] - g_k\|_2^2 + \mathbb{E} \|\hat{g}_k(u) - \mathbb{E}[\hat{g}_k(u)]\|_2^2. \end{aligned} \quad (15)$$

The two terms are squared bias and sampling variation. Since the IS estimator is unbiased, a controlled rule has lower MSE exactly when its reduction in sampling variation exceeds the squared bias introduced by changing the coefficients. We call this the estimator-level bias–variance crossover. The next theorem gives the expected-improvement certificate used to compare such estimators at a common fixed step.

Theorem 4 (Improvement certificate). *Suppose that J is L_k -smooth on every update segment generated by a coefficient rule u almost surely, for a constant $L_k > 0$, and that $\mathbb{E} \|\hat{g}_k(u)\|_2^2 < \infty$. For any step size $\eta_k > 0$,*

$$\mathbb{E} [J(\theta_k + \eta_k \hat{g}_k(u)) - J(\theta_k)] \geq \eta_k g_k^\top \mathbb{E}[\hat{g}_k(u)] - \frac{L_k \eta_k^2}{2} \mathbb{E} \|\hat{g}_k(u)\|_2^2. \quad (16)$$

Denote the right-hand side by

$$B_k(u; \eta_k) := \eta_k g_k^\top \mathbb{E}[\hat{g}_k(u)] - \frac{L_k \eta_k^2}{2} \mathbb{E} \|\hat{g}_k(u)\|_2^2. \quad (17)$$

The certificate has two parts. The first rewards alignment with the population gradient; the second penalizes the expected squared update under smoothness. Coefficient control improves the certificate only when the penalty it removes outweighs the useful alignment it removes. The next corollary states this crossover directly.

Corollary 5 (IS-to-controlled-estimator crossover). *Let u_{IS} denote the IS coefficient rule and let u be any alternative rule. Suppose the same deterministic L_k is valid almost surely for both families of update segments. With a common step size $0 < \eta_k \leq 1/L_k$, the alternative rule has the stronger certificate if and only if*

$$B_k(u; \eta_k) > B_k(u_{\text{IS}}; \eta_k) \iff m_k(u_{\text{IS}}) - m_k(u) > (1 - L_k \eta_k) \{ \mathbb{E} \|\hat{g}_k(u_{\text{IS}})\|_2^2 - \mathbb{E} \|\hat{g}_k(u)\|_2^2 \}. \quad (18)$$

When the alternative lowers the update second moment, the right-hand side of Equation (18) is its discounted reduction. Its MSE gain must exceed this amount. Since the IS estimator is unbiased, the same condition has the equivalent alignment form

$$\|g_k\|_2^2 - g_k^\top \mathbb{E}[\hat{g}_k(u)] < \frac{L_k \eta_k}{2} \{ \mathbb{E}\|\hat{g}_k(u_{\text{IS}})\|_2^2 - \mathbb{E}\|\hat{g}_k(u)\|_2^2 \}. \quad (19)$$

The left-hand side is the population-gradient alignment lost by changing the estimator; the right-hand side is the reduction in the quadratic smoothness penalty.

Remark 6 (Truncated importance sampling). Upper truncation, $u_{\text{T},t} = \min\{r_t, c\}$, caps each selected ratio without deleting its tail contribution. It is therefore a cheap variance safeguard with comparatively mild bias intervention. Its control is also limited: it acts only on the chosen upper tail, so the reduction in the update second moment can be modest under sequence-level mismatch.

Remark 7 (PPO-style masking). The PPO rule in Equation (6) removes saturated positive-advantage terms and low-ratio negative-advantage terms. It can reduce the update second moment more aggressively than upper truncation, but it also discards more population-gradient alignment. Equations (18) and (19) state when that stronger intervention is worthwhile.

5 Policy optimization and ESS

This section makes the theoretical distinction observable in two settings. An exactly solvable Gaussian bandit first separates gradient-estimation MSE from one-step policy improvement. Large-model trajectories then show how observed effective support evolves around delayed optimization failure.

5.1 Policy optimization in a Gaussian bandit

To isolate the estimator and optimization comparisons, we construct a scalar Gaussian bandit with a quadratic reward. Gaussian policies and smooth objectives provide a standard analytic setting for policy-gradient methods [5, 8, 10]. We hold the current policy fixed and vary only its distance from the rollout policy. Along this path, every quantity is available in closed form and contains no Monte Carlo error; Appendix A.1 gives the setup and calculations.

At each point on this path, the MSE oracle selects between IS and PPO by gradient-estimation MSE, whereas the certificate oracle selects the candidate with larger $B_k(u; \eta_k)$. The safe certificate oracle additionally includes a null update with certificate zero. We use $N = 32$, PPO radius $\epsilon = 0.2$, gradient $g = 2$, and step size $\eta = 0.4$, and vary ρ along one monotone ESS path.

Figure 2 first shows why MSE matters. As effective support contracts from $\rho = 0.961$ to 0.0109, the IS MSE rises from 0.228 to 62.3 and its exact expected improvement falls from 1.26 to -3.70 . The same ESS path therefore reveals the reliability loss before choosing a particular intervention. It is a regime diagnostic, not the oracle: the pairwise IS–PPO MSE crossover occurs at $\rho_{\text{MSE}} = 0.123$, while their certificate crossover occurs later at $\rho_B = 0.0392$.

The gap between the two crossovers shows why MSE alone does not determine the better update. PPO has lower MSE below ρ_{MSE} , but masking also removes population-gradient alignment. Its certificate therefore overtakes IS only below ρ_B , where the reduction in update variability finally compensates for the discarded signal. At still lower support, PPO reaches zero certificate at $\rho_{\text{stop}} = 0.0109$ and the safe oracle abstains. These values are specific to (g, N, ϵ, η) , but the ordering is the point: ESS locates the support regime, while MSE reduction and retained alignment jointly determine the better update.

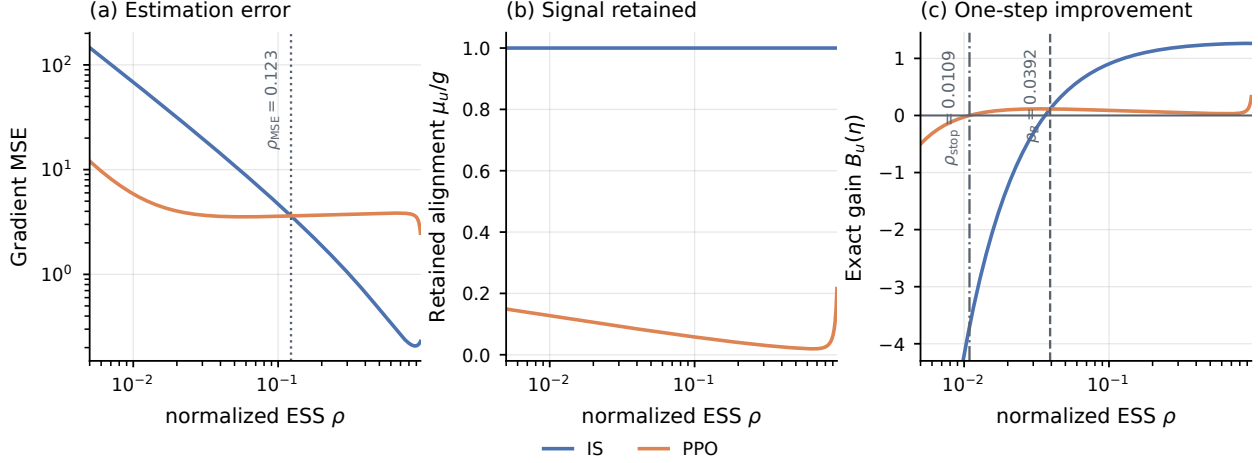


Figure 2: Exact Gaussian-bandit comparison with $g = 2$, $N = 32$, $\epsilon = 0.2$, and $\eta = 0.4$. (a) Gradient MSE as effective support contracts. (b) Population-gradient alignment retained by each estimator. (c) Exact expected one-step improvement, equal here to $B_u(\eta)$. The dotted line in panel (a) marks $\rho_{\text{MSE}} = 0.123$; in panel (c), the dashed line marks $\rho_B = 0.0392$ and the dash-dotted line marks PPO’s zero certificate at $\rho_{\text{stop}} = 0.0109$.

5.2 Policy optimization in large language models

The Gaussian comparison holds the policy state fixed. Large-model training adds a temporal question: does observed effective support deteriorate before the performance loss becomes visible? Figure 3 places the two permissive Qwen3-30B-A3B trajectories from Figure 1, which use TIS caps 3 and 5, beside a third trajectory trained with GRPO without clipping. Each run is trained for up to 200 policy updates and evaluated on AIME-2024 using mean@16.

Persistent support loss accompanies the loss of further policy improvement in both TIS trajectories. After both trajectories peak at step 60, TIS 3 and TIS 5 begin runs of at least five consecutive sub-0.01 ESS observations at steps 80 and 68, respectively, and both later collapse. The unclipped trajectory clarifies why persistence matters: its sub-0.01 observation at step 57 is isolated, ESS rebounds immediately, and AIME performance continues from 28.1% at step 60 to a 44.4% peak at step 140. Thus, in these trajectories, sustained near-zero ESS is associated with the loss of further policy improvement, whereas one noisy crossing is not. ESS supplies the warning signal; the controlled estimator determines the low-support action.

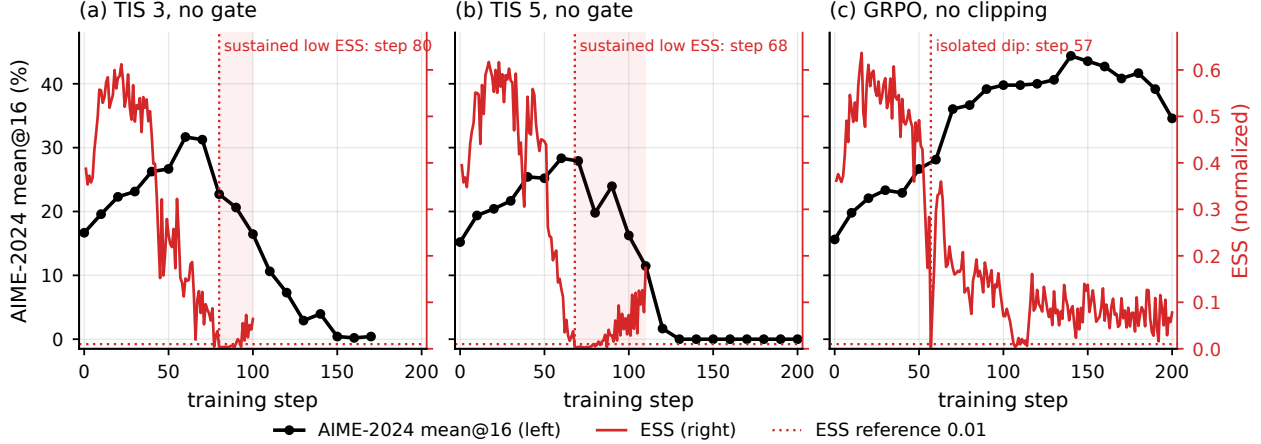


Figure 3: Effective support along three permissive Qwen3-30B-A3B trajectories. Panels (a) and (b) reproduce the TIS-3 and TIS-5 motivation runs. Their sustained low-ESS episodes, defined descriptively as five consecutive observations below 0.01, begin at steps 80 and 68 and are followed by continued deterioration. Panel (c) shows GRPO without clipping: its crossing at step 57 is isolated, ESS rebounds, and performance continues improving to step 140. Black curves report AIME-2024 mean@16 and red curves report normalized sequence ESS. The horizontal dotted line marks 0.01; vertical dotted lines mark the sustained onset in panels (a)–(b) and the isolated dip in panel (c). Shading extends to the first post-peak evaluation below initial performance, where each TIS ESS trace is stopped because post-collapse ESS no longer diagnoses the failed update path.

The two examples give ESS a precise empirical role. In the Gaussian bandit, ESS indexes the support regime, while the full certificate determines the preferred update. In the language-model trajectories, sample ESS reveals persistent support loss before failure becomes visible. The operational question is therefore whether conditioning a safeguard on ESS improves low-support updates without sacrificing permissive learning when support is high. We test this rule next.

6 ESS-conditioned policy optimization

Section 5 identifies ESS as a coordinate for effective support. A natural extension is to test whether this coordinate can time an existing safeguard: retain a permissive update while observed support is broad, and activate control only after support contracts. We first study this transition in Qwen3-30B-A3B math RLVR, then change the model scale and training objective.

6.1 Update rule and protocol

Let $\hat{g}_{\text{perm}}$ denote the permissive update and $\hat{g}_{\text{safe}}$ a prespecified safeguard. Given a threshold τ , the ESS-conditioned update is

$$\hat{g}_{\text{gate}} = \begin{cases} \hat{g}_{\text{perm}}, & \hat{\rho} \geq \tau, \\ \hat{g}_{\text{safe}}, & \hat{\rho} < \tau. \end{cases} \quad (20)$$

ESS determines when the update changes; the selected safeguard determines how the low-support update is controlled. We fix $\tau = 0.1$ throughout. This value was prescribed for the language-model runs and is not derived from the Gaussian thresholds in Section 5.1.

The main experiment trains Qwen3-30B-A3B for 200 policy updates on the boxed mathematical-reasoning data of Yu et al. [18] and evaluates AIME-2024 using mean@16. The permissive branch uses truncated importance sampling (TIS) with ratio cap 3, or no coefficient clipping in the explicit unclipped ablation. The low-support branch uses conventional GRPO clipping, Clip-Higher [18], DPPO [11], or skips the minibatch update. Most variants compute $\hat{\rho}$ from the raw complete-sequence ratios before coefficient control. The 30B TIS/Clip-Higher and TIS-5 constructions, their 8B counterpart, and the RLHF transfer instead compute the same statistic after the coefficient cap; we refer to this implementation as shaped ESS. Results are complete training trajectories, so the comparisons below describe the temporal and endpoint behavior of the reported runs. Each curve represents one training trajectory; no across-seed uncertainty is available.

6.2 Main results

Figure 4 tests one conditional template: use TIS 3 while effective support is high, then activate a prespecified clipping rule when ESS falls below 0.1. Panels (a)–(c) vary the low-support safeguard and compare the conditional construction with its always-on counterpart. Panel (d) directly exposes the targeted failure: TIS 3 alone learns quickly and then collapses, whereas ESS-conditioned clipping preserves the gain. The first three panels are complete-recipe comparisons matched on the low-support safeguard.

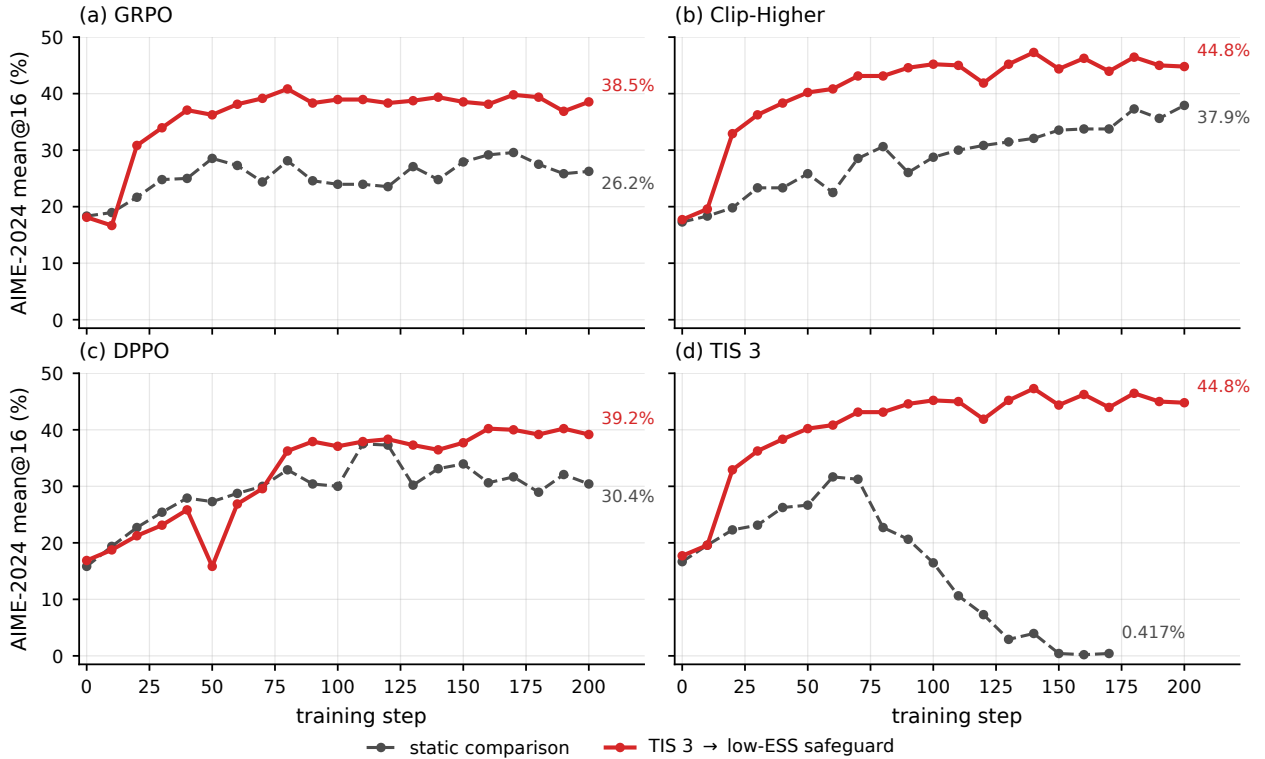


Figure 4: ESS-conditioned policy optimization on Qwen3-30B-A3B. (a) Static GRPO versus TIS 3 at high ESS and GRPO clipping at low ESS. (b) Static Clip-Higher versus TIS 3 at high ESS and Clip-Higher at low ESS. (c) Static DPPO versus TIS 3 at high ESS and DPPO at low ESS. (d) Static TIS 3 versus the TIS-3/Clip-Higher trajectory from panel (b). Direct labels give the last reported AIME-2024 mean@16; static TIS 3 ends at step 170 after collapse, while the other trajectories end at step 200.

GRPO. Static GRPO peaks at 29.6% and finishes at 26.2%. The conditional recipe, which uses TIS 3 while ESS is high and the standard 0.2/0.2 GRPO band when ESS is low, peaks at 40.8% and finishes at 38.5%. Selective clipping therefore retains the permissive high-support phase while improving both the peak and the final evaluation over always-on GRPO.

Clip-Higher. Always applying the 0.2/0.28 Clip-Higher band reaches and finishes at 37.9%. Using TIS 3 at high ESS and activating the same band only at low ESS instead peaks at 47.3% and finishes at 44.8%. Delaying the safeguard until support contracts again improves the entire observed trajectory.

DPPO. Static DPPO peaks at 37.5% but recedes to 30.4%, whereas the TIS-3/DPPO conditional construction peaks at 40.2% and finishes at 39.2%. The same high-ESS TIS-3 and low-ESS safeguard pattern therefore preserves late performance under a third clipping rule.

TIS 3. The direct TIS-3 comparison makes the failure-prevention effect explicit. Without ESS conditioning, TIS 3 reaches 31.7% at step 60 and then collapses to 0.417% at its last recorded evaluation, step 170. Switching to Clip-Higher only in the low-ESS regime instead reaches 47.3% and finishes at 44.8%. ESS conditioning prevents the observed permissive-estimator collapse while retaining its high-support learning phase.

Unclipped update. The unclipped comparison tests whether the same timing principle extends beyond TIS. Both runs in Figure 5 use an unclipped high-support update, but one activates conventional clipping when raw ESS is below 0.1. The unconditioned trajectory peaks at 44.4% and finishes at 34.6%; conditional clipping retains a 43.3% peak and improves the endpoint to 38.1%. Low-ESS clipping therefore preserves most of the permissive early gain while improving late retention.

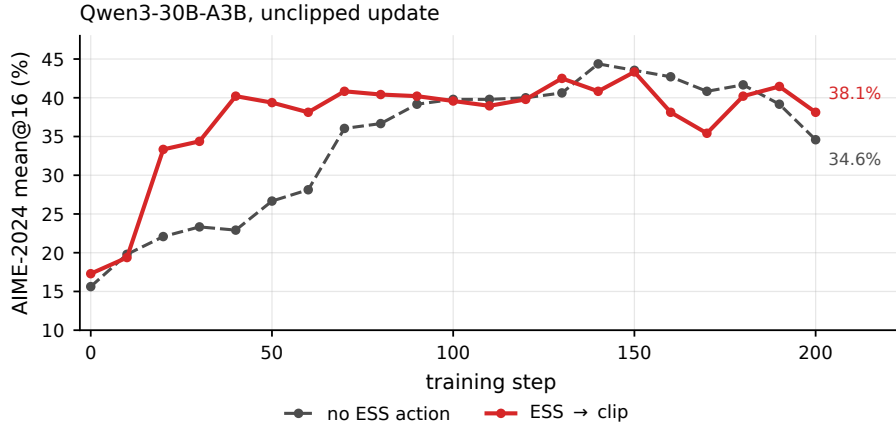


Figure 5: ESS-conditioned clipping with an unclipped Qwen3-30B-A3B update. The comparison activates conventional clipping only when raw normalized ESS is below 0.1. The reference has no ESS-conditioned action; its nominal fraction trigger produces only three small interventions. Direct labels give the step-200 AIME-2024 mean@16.

The implemented intervention also follows the predicted temporal ordering. For the DPPO construction, mean raw ESS falls from 0.412 over steps 1–50 to 0.120 over steps 151–200, while the mean intervention fraction rises from 0.153 to 0.668. Figure 9 reports the corresponding patterns

for all three safeguards. Protection is therefore present but limited early, and intensifies as effective support contracts.

Other ESS-conditioned optimization variants. A hard update veto provides a simpler low-support action. Figure 6 groups this alternative across two permissive updates. With an unclipped high-support update, ESS-conditioned skipping finishes at 35.6%, compared with 34.6% without an ESS action and 38.1% under conditional clipping. The matched TIS-5 pair gives the sharper failure-prevention result: static TIS 5 peaks at 28.3% and collapses to 0.00%, whereas the trajectory that skips low-ESS updates reaches and finishes at 39.4%. The same observed collapse is therefore prevented at a second TIS cap and with a simpler ESS action, although skipping need not be the best action when the permissive run remains viable.

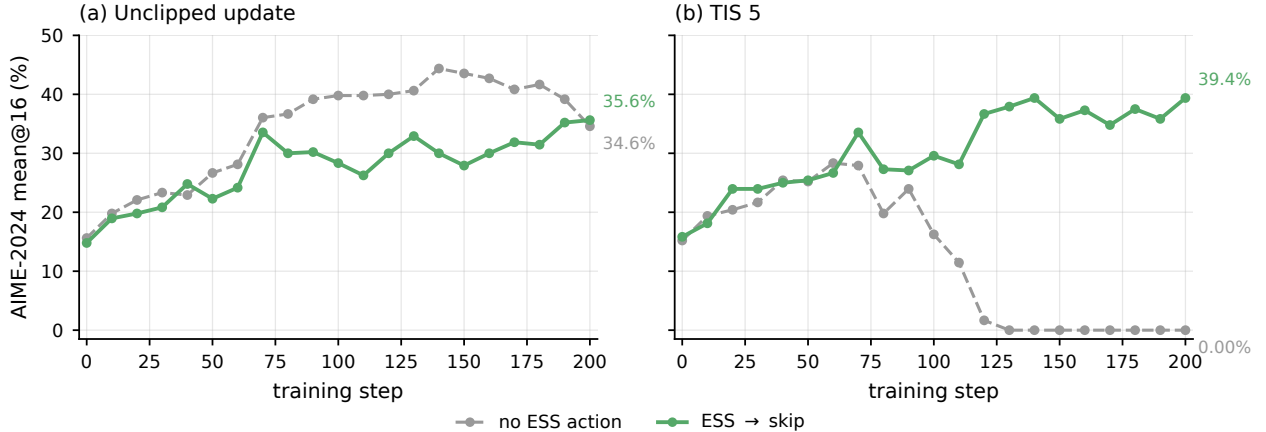


Figure 6: ESS-conditioned skipping as an alternative low-support action. Panel (a) applies the veto to an otherwise unclipped update. Panel (b) gives the matched TIS-5 comparison: static TIS 5 collapses, whereas skipping low-support updates finishes at 39.4%. All curves report AIME-2024 mean@16.

6.3 Transfer across models and objectives

We first change the model scale while keeping the math-RLVR task fixed. Figure 7 reports the two Qwen3-8B comparisons for which loss aggregation is matched. Under Clip-Higher, the static trajectory finishes at 29.0%, while the ESS-conditioned construction finishes at 32.7%. Under DPPO, the corresponding endpoints are 31.7% and 32.5%. These gains are smaller than the 30B rescue and depend on the safeguard. The endpoint ordering favors the conditional construction in both aggregation-matched comparisons after the change in model scale.

Finally, we change both the objective and the data. We train Qwen3-4B-Instruct-2507 for 200 updates on Anthropic HH-RLHF [1] using Skywork-Reward-Llama-3.1-8B [4] as the reward model. Figure 8 compares the complete GRPO and TIS-3-plus-ESS recipes. Their reward-model scores start at comparable levels and finish at 43.0 and 60.7, respectively. This experiment tests optimization transfer on the training reward model; it does not measure human preference directly.

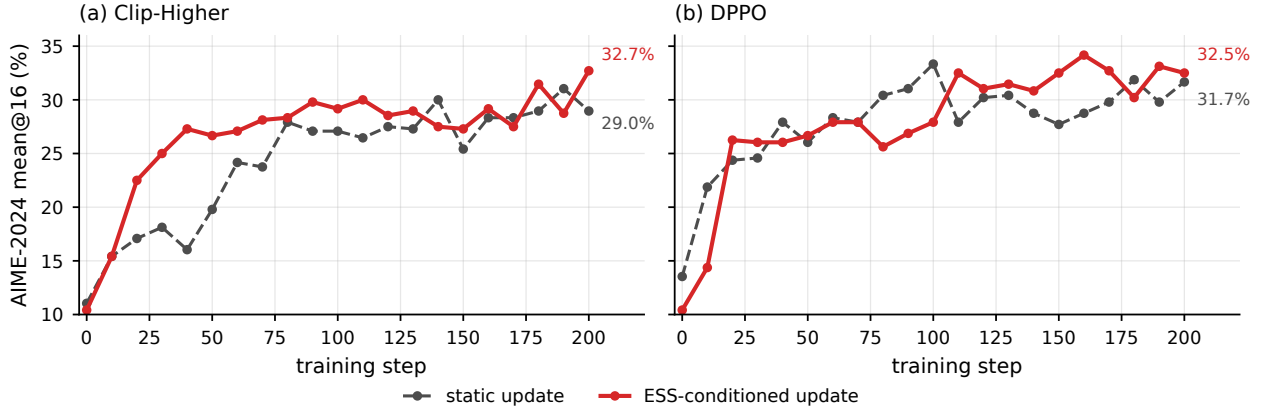


Figure 7: Dense-model transfer on Qwen3-8B. Panel (a) compares token-mean Clip-Higher with its token-mean ESS-conditioned construction. Panel (b) compares static and ESS-conditioned DPPO under the same sum-normalized aggregation. Direct labels give the step-200 AIME-2024 mean@16.

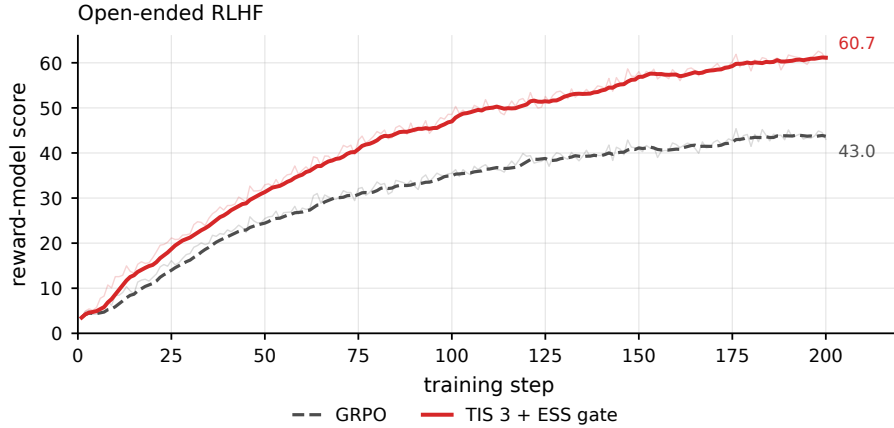


Figure 8: Transfer to open-ended RLHF on Qwen3-4B-Instruct-2507. Thin curves are per-step reward-model scores; thick curves are trailing seven-step means. Direct labels give the step-200 observations. This is a complete-recipe comparison between GRPO and TIS 3 with an ESS-conditioned safeguard.

Across these experiments, ESS contributes timing rather than a new clipping rule. High observed support leaves the permissive branch active; lower support increases the use of a prespecified control. The 30B rescue patterns, the smaller 8B gains, and the RLHF transfer are consistent with the same division of labor: ESS locates the reliability regime, while the safeguarded estimator determines the low-support update.

7 Conclusion

We studied clipping as a finite-sample estimator choice rather than a default property of policy optimization. The theory separates gradient-estimation error from the useful update retained after control: reducing MSE improves the fixed-step certificate only when that gain compensates for the signal removed by clipping. This distinction makes IS preferable while effective support is broad and explains why controlled estimators become attractive as mismatch grows. The exact Gaussian construction exhibits separate MSE and update crossovers. The reported language-model trajectories then show that sequence ESS can reveal support contraction and time a prespecified safeguard without serving as the oracle itself. In the RLVR and RLHF examples, the resulting conditional constructions retain permissive learning in the higher-support regime and protect training more strongly as support contracts. ESS therefore provides the operational link between importance-weight reliability and the decision of when to control an update.

A Numerical Experiments

A.1 Gaussian bandit

This section gives the calculations used in Section 5.1.

Setup We compare IS and the PPO-style rule in Equation (6), using the exact current-policy value baseline.

Because the score contribution is polynomial and each mask boundary is linear in z , the means, MSEs, and update second moments are finite sums of truncated Gaussian moments.

We choose a quadratic reward because it makes the smoothness certificate equal to the true expected gain. Any separation between the two oracles is therefore genuine rather than an artifact of a loose bound.

Consider the current policy and rollout policy

$$P_\theta = \mathcal{N}(\theta, 1), \quad Q_\Delta = \mathcal{N}(\theta - \Delta, 1), \quad \Delta \geq 0, \quad (21)$$

with reward $R(a) = -\frac{1}{2}(a - a^*)^2$. We hold the current policy fixed and vary only the rollout gap Δ . Writing $g := a^* - \theta > 0$, the objective is

$$J(\theta) = -\frac{1}{2}\{(\theta - a^*)^2 + 1\}, \quad \nabla J(\theta) = g, \quad L = 1. \quad (22)$$

Here θ is the fixed pre-update mean and Q_Δ remains frozen during the hypothetical one-step update. We vary Δ only across common-state comparisons. For $z = a - \theta$, the likelihood ratio and normalized ESS reduce to

$$W_\Delta(z) = \exp\{\Delta z + \Delta^2/2\}, \quad \rho(\Delta) = \{\mathbb{E}_{Q_\Delta}[W_\Delta^2]\}^{-1} = \exp(-\Delta^2). \quad (23)$$

Normalized ESS is therefore a one-to-one coordinate for this branch.

Write $\mu_u := \mathbb{E}[\hat{g}_u]$, $s_u := \mathbb{E}[\hat{g}_u^2]$, and $B_u(\eta) := \eta g \mu_u - \eta^2 s_u / 2$. For this quadratic objective, the fixed-step certificate is exact:

$$\mathbb{E}[J(\theta + \eta \hat{g}_u) - J(\theta)] = \eta g \mu_u - \frac{\eta^2}{2} s_u = B_u(\eta). \quad (24)$$

We use $N = 32$, the standard PPO radius $\epsilon = 0.2$, and choose $g = 2$ and $\eta = 0.4$ so that the MSE, improvement, and abstention regimes are all visible on one monotone ESS path.

Exact Closed Form Computations Let $z = a - \theta$. Under the current policy, $z \sim \mathcal{N}(0, 1)$ and the score is z . The exact current-policy value baseline gives

$$A_g(z) = gz + \frac{1 - z^2}{2}, \quad f_g(z) := z A_g(z) = gz^2 + \frac{z - z^3}{2}. \quad (25)$$

Although f_g is unbounded, all polynomial Gaussian moments used below are finite. The example therefore uses the finite-moment fixed-step certificate; the bounded-integrand IS theorem addresses a separate reliability question. IS uses $W_\Delta f_g$. The PPO-style candidate uses $W_\Delta f_g M_{\epsilon, \Delta}$, where

$$M_{\epsilon, \Delta}(z) := \mathbf{1}\{A_g(z) \geq 0, W_\Delta(z) \leq 1 + \epsilon\} + \mathbf{1}\{A_g(z) < 0, W_\Delta(z) \geq 1 - \epsilon\}. \quad (26)$$

Let $M_{\text{IS}} \equiv 1$ and $M_{\text{P}} = M_{\epsilon, \Delta}$. Change of measure gives the exact per-sample moments

$$\begin{aligned} \mu_u(\Delta) &= \mathbb{E}_{U \sim \mathcal{N}(0, 1)}[f_g(U) M_u(U)], \\ h_u(\Delta) &= e^{\Delta^2} \mathbb{E}_{V \sim \mathcal{N}(\Delta, 1)}[f_g(V)^2 M_u(V)], \quad u \in \{\text{IS}, \text{P}\}. \end{aligned} \quad (27)$$

In particular, $\mu_{\text{IS}} = g$. For an iid estimator with batch size N ,

$$m_u = (\mu_u - g)^2 + \frac{h_u - \mu_u^2}{N}, \quad s_u = \mu_u^2 + \frac{h_u - \mu_u^2}{N}. \quad (28)$$

The advantage changes sign at $g \pm \sqrt{g^2 + 1}$, and the ratio boundaries are $(\log(1 \pm \epsilon) - \Delta^2/2)/\Delta$. Equation (27) is therefore a finite sum of truncated normal polynomial moments. With $I_j(a, b) := \int_a^b z^j \phi(z) dz$, integration by parts gives

$$I_j(a, b) = a^{j-1} \phi(a) - b^{j-1} \phi(b) + (j-1) I_{j-2}(a, b), \quad j \geq 2, \quad (29)$$

starting from $I_0 = \Phi(b) - \Phi(a)$ and $I_1 = \phi(a) - \phi(b)$. Boundary terms vanish at infinity. Substituting these moments into Equation (28) and the exact gain in Equation (24) gives the reported boundaries.

Results We evaluate $0.2 \leq \Delta \leq \sqrt{-\log(0.005)}$. Numerical evaluation of the closed-form moments gives three substantive boundaries on this branch:

$$\rho_{\text{MSE}} = 0.123, \quad \rho_B = 0.0392, \quad \rho_{\text{stop}} = 0.0109. \quad (30)$$

The MSE oracle over $\mathcal{U} = \{u_{\text{IS}}, u_{\text{P}}\}$ selects IS above ρ_{MSE} and PPO below it. The safe certificate oracle adds the null update and instead selects IS for $\rho > \rho_B$, PPO for $\rho_{\text{stop}} < \rho < \rho_B$, and no update for $\rho < \rho_{\text{stop}}$. Adjacent choices tie at each boundary.

B Additional coefficient-rule details

The main text treats coefficient control generically because different rules alter different parts of the gradient. A factorwise detached upper truncation uses $u_{T,t}(\mathbf{r}, A) := \min\{r_t, c\}$. Its contribution obeys

$$\|\psi_k(u_T; Z)\|_2 \leq c|A(Z)| \sum_{t=1}^T \|\ell_{\theta_k,t}(Z)\|_2, \quad (31)$$

so it has a direct second-moment envelope whenever the right-hand side has a finite second moment.

PPO masking is different. For $A \geq 0$ and $c = 1 + \epsilon$,

$$0 \leq r - \min\{r, c\} = (r - c)_+ \leq r \mathbf{1}\{r > c\} = r - r M_\epsilon(r, A), \quad (32)$$

where $M_\epsilon(r, A) = \mathbf{1}\{A \geq 0, r \leq 1 + \epsilon\} + \mathbf{1}\{A < 0, r \geq 1 - \epsilon\}$. Truncation retains a bounded positive coefficient above the cap, whereas PPO removes that contribution. On the negative-advantage lower branch, PPO masks small ratios whereas upper truncation does not. A complete-sample mask is another distinct rule: it sets $u_t(\mathbf{r}, A) = W_k M_\epsilon(W_k, A)$ for every factor. We state which rule is used in each experiment rather than treating these implementations as interchangeable.

C Additional RLVR diagnostics

C.1 Intervention and training dynamics

The main text reports evaluation performance. We record here how often the conditional rule intervenes and the accompanying training dynamics.

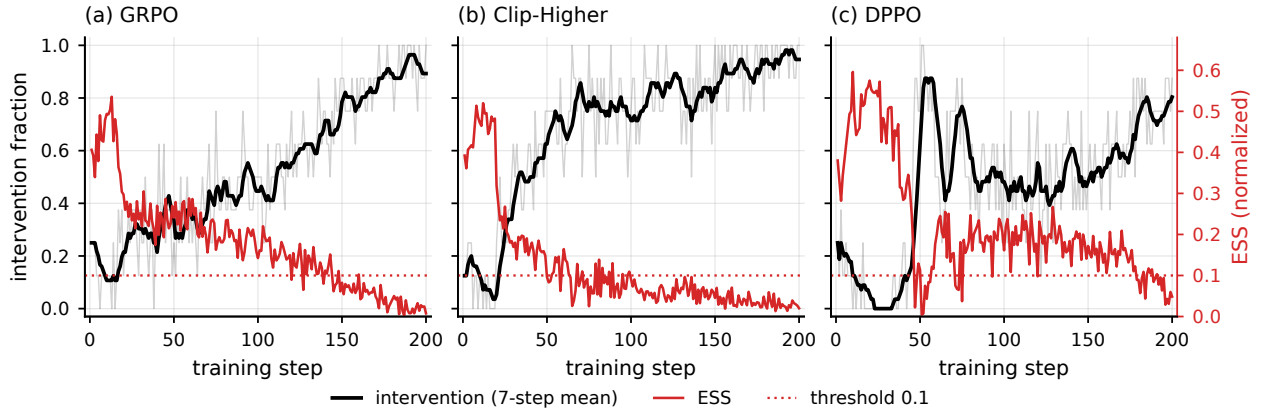


Figure 9: Realized ESS-conditioned intervention schedules. Each panel pairs the per-step intervention fraction (thin black) and its trailing seven-step mean (thick black) with normalized ESS (red). GRPO and DPPO use raw ESS; Clip-Higher uses shaped ESS after the coefficient cap. The dotted line marks the threshold 0.1.

Figure 9 shows the intended temporal ordering: intervention is already possible early, but its frequency increases as ESS contracts. The remaining figures record response length, entropy, and reward diagnostics.

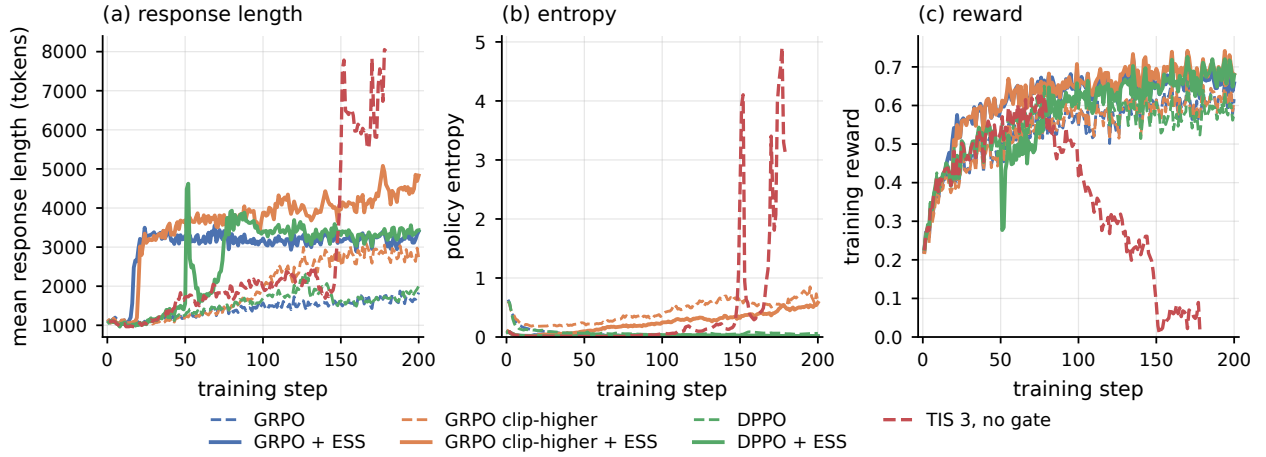


Figure 10: Training dynamics for the main comparisons. The static TIS 3 run develops extreme response length and entropy together with a collapse in training reward. The safeguards and their ESS-conditional variants avoid this failure over the reported 200 steps.

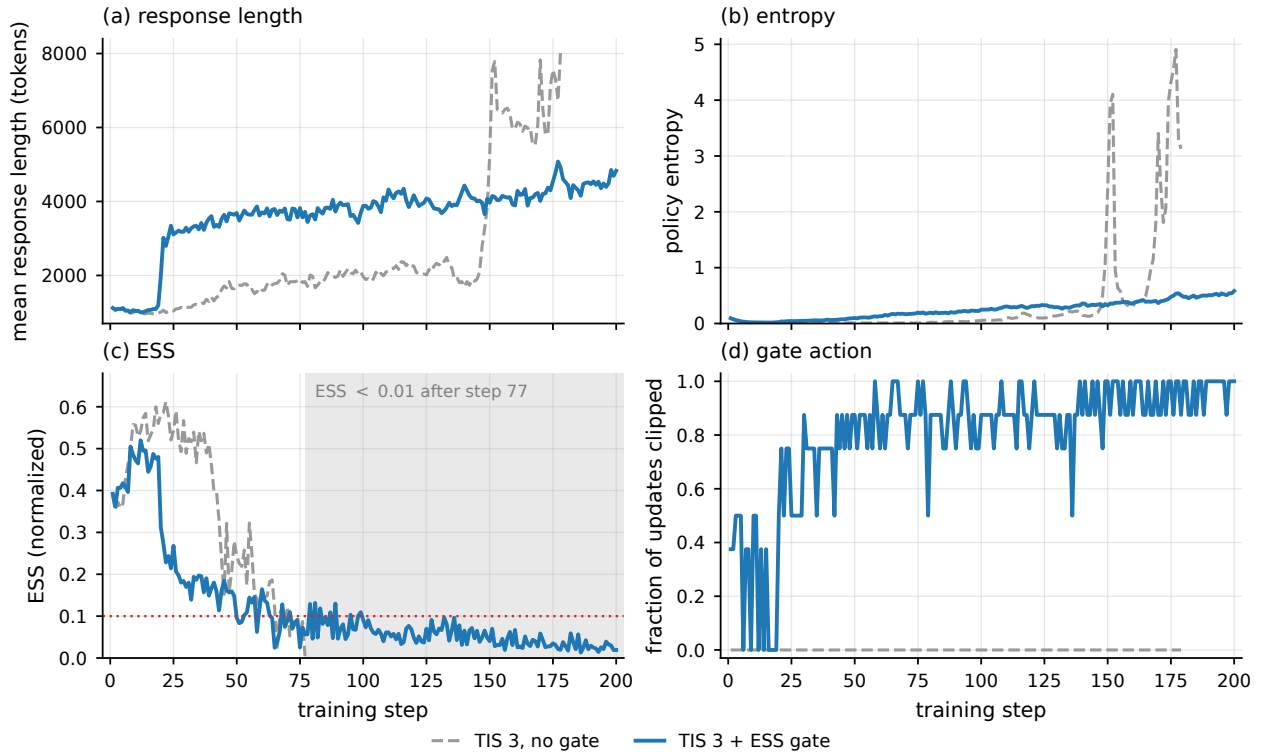


Figure 11: Detailed TIS 3 comparison. Under static TIS, ESS falls below 0.01 after step 77 and the response length and entropy diverge. The fixed-0.1 intervention increases the clipped fraction as ESS declines and keeps both diagnostics controlled.

D Proofs for the theoretical results

At update k , all expectations in this appendix are taken after conditioning on the preceding updates. The current policy is therefore fixed and the unused sampling units are independent draws from Q . We suppress this conditioning throughout.

D.1 Policy-gradient identity and mean-square bound

By definition, $f_k = (R - b(X))\nabla_\theta \log P_{\theta_k}(Y | X)$ and $W_k = dP_{\theta_k}/dQ$. Change of measure gives

$$\begin{aligned}\mathbb{E}_Q[W_k f_k] &= \mathbb{E}_{P_{\theta_k}}[(R - b(X))\nabla_\theta \log P_{\theta_k}(Y | X)] \\ &= \mathbb{E}_{P_{\theta_k}}[R\nabla_\theta \log P_{\theta_k}(Y | X)] \\ &\quad - \mathbb{E}_{P_{\theta_k}}[b(X)\nabla_\theta \log P_{\theta_k}(Y | X)].\end{aligned}\tag{33}$$

For the baseline term, condition further on X . Under the standard regularity condition that permits differentiation under the conditional integral,

$$\begin{aligned}\mathbb{E}_{P_{\theta_k}}[b(X)\nabla_\theta \log P_{\theta_k}(Y | X) | X] &= b(X) \int P_{\theta_k}(y | X) \nabla_\theta \log P_{\theta_k}(y | X) d\lambda(y) \\ &= b(X) \nabla_\theta \int P_{\theta_k}(y | X) d\lambda(y) \\ &= b(X) \nabla_\theta 1 = 0.\end{aligned}\tag{34}$$

The remaining term is the score-function identity. Hence

$$\mathbb{E}_Q[W_k f_k] = \nabla J(\theta_k) = g_k.\tag{35}$$

Averaging N independent copies gives $\mathbb{E}[\hat{g}_k] = g_k$.

For the mean-square calculation, define $\xi_{k,i} := W_{k,i} f_{k,i} - g_k$. Equation (35) implies $\mathbb{E}[\xi_{k,i}] = 0$. Since $\hat{g}_k - g_k = N^{-1} \sum_{i=1}^N \xi_{k,i}$,

$$\mathbb{E}\|\hat{g}_k - g_k\|_2^2 = \frac{1}{N^2} \sum_{i=1}^N \mathbb{E}\|\xi_{k,i}\|_2^2 + \frac{1}{N^2} \sum_{i \neq j} \mathbb{E}[\xi_{k,i}^\top \xi_{k,j}].\tag{36}$$

For $i \neq j$, independence and zero means give

$$\mathbb{E}[\xi_{k,i}^\top \xi_{k,j}] = \mathbb{E}[\xi_{k,i}]^\top \mathbb{E}[\xi_{k,j}] = 0.\tag{37}$$

The diagonal terms are identical, so

$$\mathbb{E}\|\hat{g}_k - g_k\|_2^2 = \frac{1}{N} \mathbb{E}_Q \|W_k f_k - g_k\|_2^2.\tag{38}$$

Expanding the remaining squared norm yields

$$\begin{aligned}\mathbb{E}_Q \|W_k f_k - g_k\|_2^2 &= \mathbb{E}_Q [W_k^2 \|f_k\|_2^2] - 2g_k^\top \mathbb{E}_Q [W_k f_k] + \|g_k\|_2^2 \\ &= \mathbb{E}_Q [W_k^2 \|f_k\|_2^2] - \|g_k\|_2^2,\end{aligned}\tag{39}$$

where the last equality uses Equation (35). Combining Equations (38) and (39) gives

$$\mathbb{E}\|\widehat{g}_k - g_k\|_2^2 = \frac{1}{N} \left\{ \mathbb{E}_Q[W_k^2 \|f_k\|_2^2] - \|g_k\|_2^2 \right\}. \quad (40)$$

Finally, $\|f_k(z)\|_2 \leq \|f_k\|_\infty$ and $\mathbb{E}_Q[W_k^2] = \rho_k^{-1}$ imply

$$\begin{aligned} \mathbb{E}\|\widehat{g}_k - g_k\|_2^2 &\leq \frac{1}{N} \mathbb{E}_Q[W_k^2 \|f_k\|_2^2] \\ &\leq \frac{\|f_k\|_\infty^2}{N} \mathbb{E}_Q[W_k^2] \\ &= \frac{\|f_k\|_\infty^2}{N\rho_k}, \end{aligned} \quad (41)$$

which proves Equation (11).

D.2 Proof of Lemma 2

Apply Markov's inequality to the nonnegative random variable $\|\widehat{g}_k - g_k\|_2^2$:

$$\begin{aligned} \Pr\left(\|\widehat{g}_k - g_k\|_2 > \frac{\|f_k\|_\infty}{\sqrt{\delta N \rho_k}}\right) &= \Pr\left(\|\widehat{g}_k - g_k\|_2^2 > \frac{\|f_k\|_\infty^2}{\delta N \rho_k}\right) \\ &\leq \frac{\mathbb{E}\|\widehat{g}_k - g_k\|_2^2}{\|f_k\|_\infty^2 / (\delta N \rho_k)} \\ &\leq \delta, \end{aligned} \quad (42)$$

where the last line uses Equation (11). Taking complements proves Equation (12).

D.3 Proof of Lemma 1

Let $\|\widehat{g} - g\|_2 \leq \varepsilon$. Smoothness gives, for every $\gamma \geq 0$,

$$J(\theta + \gamma \widehat{g}) - J(\theta) \geq \gamma g^\top \widehat{g} - \frac{L\gamma^2}{2} \|\widehat{g}\|_2^2. \quad (43)$$

The inner product satisfies

$$\begin{aligned} g^\top \widehat{g} &= \|\widehat{g}\|_2^2 + (g - \widehat{g})^\top \widehat{g} \\ &\geq \|\widehat{g}\|_2^2 - \|g - \widehat{g}\|_2 \|\widehat{g}\|_2 \\ &\geq \|\widehat{g}\|_2 (\|\widehat{g}\|_2 - \varepsilon). \end{aligned} \quad (44)$$

Substituting Equation (44) into Equation (43) gives

$$J(\theta + \gamma \widehat{g}) - J(\theta) \geq \gamma \|\widehat{g}\|_2 (\|\widehat{g}\|_2 - \varepsilon) - \frac{L\gamma^2}{2} \|\widehat{g}\|_2^2. \quad (45)$$

For $\widehat{g} \neq 0$, the derivative of the right-hand side is

$$\|\widehat{g}\|_2 (\|\widehat{g}\|_2 - \varepsilon) - L\gamma \|\widehat{g}\|_2^2. \quad (46)$$

Its nonnegative maximizer is $\gamma = L^{-1}(1 - \varepsilon/\|\widehat{g}\|_2)_+$. When $\widehat{g} = 0$, both the chosen step and the claimed lower bound are zero. Substituting the maximizer into Equation (45) proves Equation (9).

D.4 Proof of Theorem 3

Lemma 2 shows that the event

$$\|\widehat{g}_k - g_k\|_2 \leq \frac{\|f_k\|_\infty}{\sqrt{\delta N \rho_k}} \quad (47)$$

has probability at least $1 - \delta$. On this event, apply Lemma 1 with $\theta = \theta_k$, $\widehat{g} = \widehat{g}_k$, $\varepsilon = \|f_k\|_\infty / \sqrt{\delta N \rho_k}$, and $L = L_k$. The resulting inequality is exactly Equation (14).

D.5 Risk and certificate comparisons for coefficient rules

For a coefficient rule u , add and subtract the estimator mean:

$$\widehat{g}_k(u) - g_k = \{\widehat{g}_k(u) - \mathbb{E}[\widehat{g}_k(u)]\} + \{\mathbb{E}[\widehat{g}_k(u)] - g_k\}. \quad (48)$$

Squaring and taking expectation gives

$$\begin{aligned} m_k(u) &= \mathbb{E} \|\widehat{g}_k(u) - \mathbb{E}[\widehat{g}_k(u)]\|_2^2 + \|\mathbb{E}[\widehat{g}_k(u)] - g_k\|_2^2 \\ &\quad + 2\{\mathbb{E}[\widehat{g}_k(u)] - g_k\}^\top \mathbb{E}[\widehat{g}_k(u) - \mathbb{E}[\widehat{g}_k(u)]] \\ &= \mathbb{E} \|\widehat{g}_k(u) - \mathbb{E}[\widehat{g}_k(u)]\|_2^2 + \|\mathbb{E}[\widehat{g}_k(u)] - g_k\|_2^2, \end{aligned} \quad (49)$$

because the centered term has mean zero. This proves Equation (15). For an iid sample mean, the first term in Equation (49) is the per-sample variation divided by N .

It remains to connect total risk to the expected policy-improvement certificate. For any random estimate $\widehat{g}$ of g_k , smoothness gives

$$J(\theta_k + \eta_k \widehat{g}) - J(\theta_k) \geq \eta_k g_k^\top \widehat{g} - \frac{L_k \eta_k^2}{2} \|\widehat{g}\|_2^2. \quad (50)$$

Taking expectation in Equation (50), with $\mathbb{E}\|\widehat{g}\|_2^2 < \infty$, gives

$$\mathbb{E}[J(\theta_k + \eta_k \widehat{g}) - J(\theta_k)] \geq \eta_k g_k^\top \mathbb{E}[\widehat{g}] - \frac{L_k \eta_k^2}{2} \mathbb{E}\|\widehat{g}\|_2^2. \quad (51)$$

Setting $\widehat{g} = \widehat{g}_k(u)$ proves Theorem 4. The polarization identity, followed by expectation, gives

$$2g_k^\top \mathbb{E}[\widehat{g}] = \|g_k\|_2^2 + \mathbb{E}\|\widehat{g}\|_2^2 - \mathbb{E}\|\widehat{g} - g_k\|_2^2. \quad (52)$$

Substituting Equation (52) into the certificate and subtracting the IS certificate gives

$$\begin{aligned} B_k(u; \eta_k) - B_k(u_{\text{IS}}; \eta_k) &= \frac{\eta_k}{2} [m_k(u_{\text{IS}}) - m_k(u)] \\ &\quad - \frac{\eta_k}{2} (1 - L_k \eta_k) \{\mathbb{E}\|\widehat{g}_k(u_{\text{IS}})\|_2^2 - \mathbb{E}\|\widehat{g}_k(u)\|_2^2\}. \end{aligned} \quad (53)$$

Since $\eta_k > 0$, the right-hand side is positive exactly under Equation (18), proving Corollary 5. Since the IS estimator is unbiased, subtracting the certificate definitions instead gives

$$\begin{aligned} B_k(u; \eta_k) - B_k(u_{\text{IS}}; \eta_k) &= -\eta_k \left\{ \|g_k\|_2^2 - g_k^\top \mathbb{E}[\widehat{g}_k(u)] \right\} \\ &\quad + \frac{L_k \eta_k^2}{2} \{\mathbb{E}\|\widehat{g}_k(u_{\text{IS}})\|_2^2 - \mathbb{E}\|\widehat{g}_k(u)\|_2^2\}. \end{aligned} \quad (54)$$

Its positivity is equivalent to Equation (19).

D.6 Intact prompt groups

Suppose M prompts are independent and each prompt has G conditionally independent responses. Keep each prompt group intact and define its contribution by $\Xi_{k,i} := G^{-1} \sum_{j=1}^G W_{k,ij} f_{k,ij}$. A leave-one-out baseline is admissible because it excludes the focal response. Conditional on the prompt and the remaining responses, the focal score still has mean zero. Hence $\mathbb{E}[\Xi_{k,i}] = g_k$.

Jensen’s inequality gives

$$\begin{aligned} \|\Xi_{k,i}\|_2^2 &= \left\| \frac{1}{G} \sum_{j=1}^G W_{k,ij} f_{k,ij} \right\|_2^2 \\ &\leq \frac{1}{G} \sum_{j=1}^G W_{k,ij}^2 \|f_{k,ij}\|_2^2. \end{aligned} \tag{55}$$

Repeating the sample-mean calculation across the M independent prompt groups gives

$$\begin{aligned} \mathbb{E} \left\| \frac{1}{M} \sum_{i=1}^M \Xi_{k,i} - g_k \right\|_2^2 &\leq \frac{1}{M} \mathbb{E} \|\Xi_{k,1}\|_2^2 \\ &\leq \frac{\|f_k\|_\infty^2}{M \rho_k}. \end{aligned} \tag{56}$$

Markov’s inequality then gives the same form as Lemma 2 with the number of independent prompts M in place of the number of responses N .

D.7 Finite-moment relaxation

The exact equality in Equation (10) requires only $\mathbb{E}_Q[W_k^2 \|f_k\|_2^2] < \infty$. The bounded-contribution condition is used solely to obtain the standard form $\|f_k\|_\infty^2 / (N \rho_k)$. More generally, define $M_{2,k} := \mathbb{E}_Q[W_k^2 \|f_k\|_2^2] / \mathbb{E}_Q[W_k^2]$. The same calculation gives $\mathbb{E} \|\hat{g}_k - g_k\|_2^2 \leq M_{2,k} / (N \rho_k)$. We use the bounded form in the main text to match conventional importance-sampling notation and make the role of normalized ESS explicit.

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